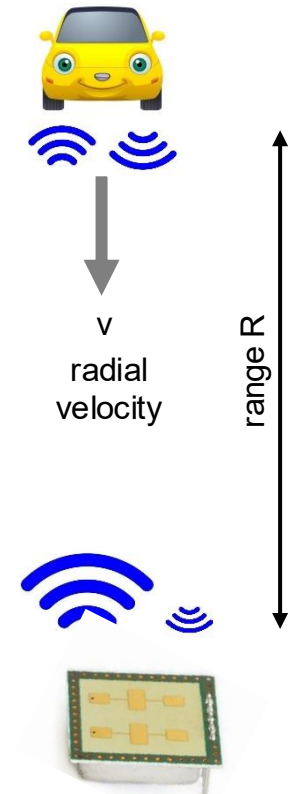


PM4 FS2026: CW-Doppler-Radar Sensor

Project Goals

- Design a radar sensor
 - start with a standard use case (see below) and/or an idea of your own
 - document your HW/DSP-concept
- Realize your prototype
 - Hardware
 - Firmware/Software
- Test the performance of your prototype
- Document and present your prototype



radar sensor
K-LC5



radar sensor
K-LC5

References

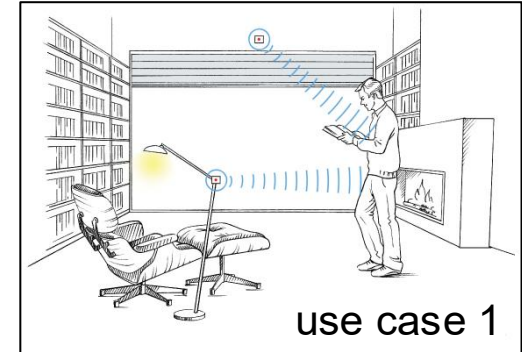
- General information can be simply found on [Wikipedia](#):
 - [Distance and Speed Measurement](#), [Radar Range Equation](#), ...
- Further technical information can be found on the [Radar Tutorial](#):
 - [Doppler-Effect](#) (see Figure 1 first)
 - [CW-Doppler-Radar](#)
(only sections Doppler Radar, Function, Block diagram of CW radar Figure 3)
 - [Radar Range Equation](#)
 - [Radar-Cross-Section](#)
- RFbeam, St. Gallen: <https://www.rfbeam.ch/>
 - [use cases](#)
 - Radar sensor used in PM4:
 - [Radar Sensor K-LC5](#), [data sheet K-LC5](#)

Some use cases

1. CW-Doppler-Radar: motion sensor

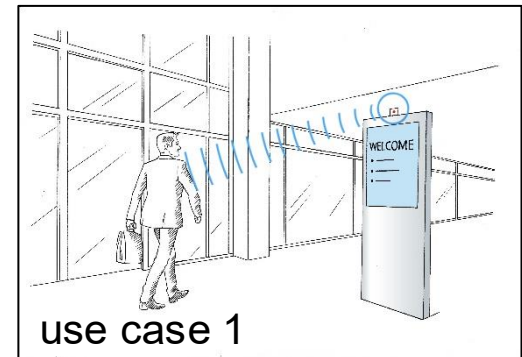
- for movement detection
- indoor automation
- intrusion alarm

challenge: human presence detection



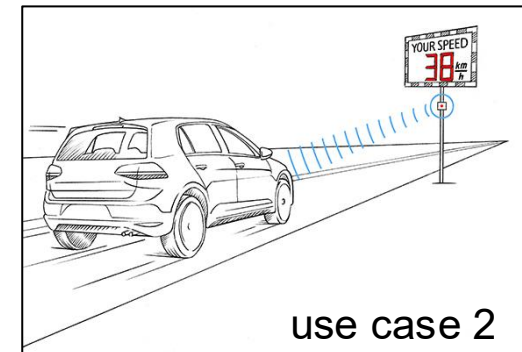
2. CW-Doppler-Radar: speed sensor

- traffic monitoring
- sports (**challenge:** reflectivity of target)



3. FMCW-Radar: range sensor

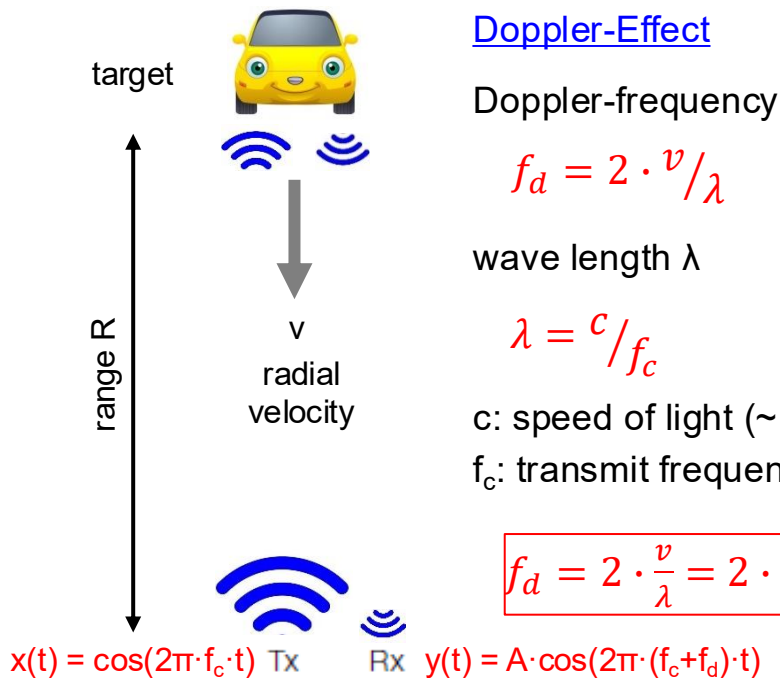
- please see corresponding document
- range (i.e. distance) measurements
- **more challenging** than CW-Doppler-Radar



Source: <https://rfbeam.ch/applications/>

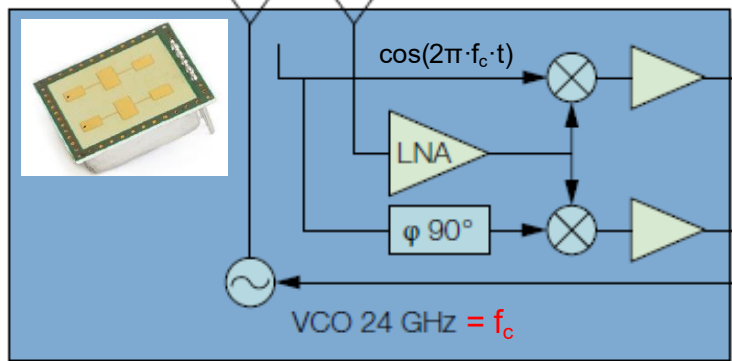
CW-Doppler-Radar: How it works

CW-Doppler-Radar



$v / (\text{m/s})$	$v / (\text{km/h})$	f_d / Hz
0.1	0.36	16
0.28	1	44.4
1	3.6	160
2.8	10	444
10	36	1600
28	100	4444

$$f_d = 2 \cdot \frac{v}{\lambda} = 2 \cdot v \cdot \frac{f_c}{c} = 2 \cdot v \cdot \frac{24 \text{ GHz}}{3 \cdot 10^8 \text{ m/s}} = \frac{v}{\text{m/s}} \cdot 160 \text{ Hz}$$



(in-phase signal) **$I(t) = A \cdot \cos(2\pi \cdot f_d \cdot t - \phi_0)$**

=> $I(t) = y(t) \cdot \cos(2\pi \cdot f_c \cdot t)$ low-pass filtered

(quadrature-phase signal) **$Q(t) = A \cdot \sin(2\pi \cdot f_d \cdot t - \phi_0)$**

=> $Q(t) = y(t) \cdot (-\sin(2\pi \cdot f_c \cdot t))$ low-pass filtered

CW-Doppler-Radar: DC to set f_c
(should be different to avoid interference between sensors)

FMCW-Radar: sawtooth
(DAC with reconstruction filter)

VCO: voltage controlled oscillator
LNA: Low Noise Amplifier

Review of Fourier Theory

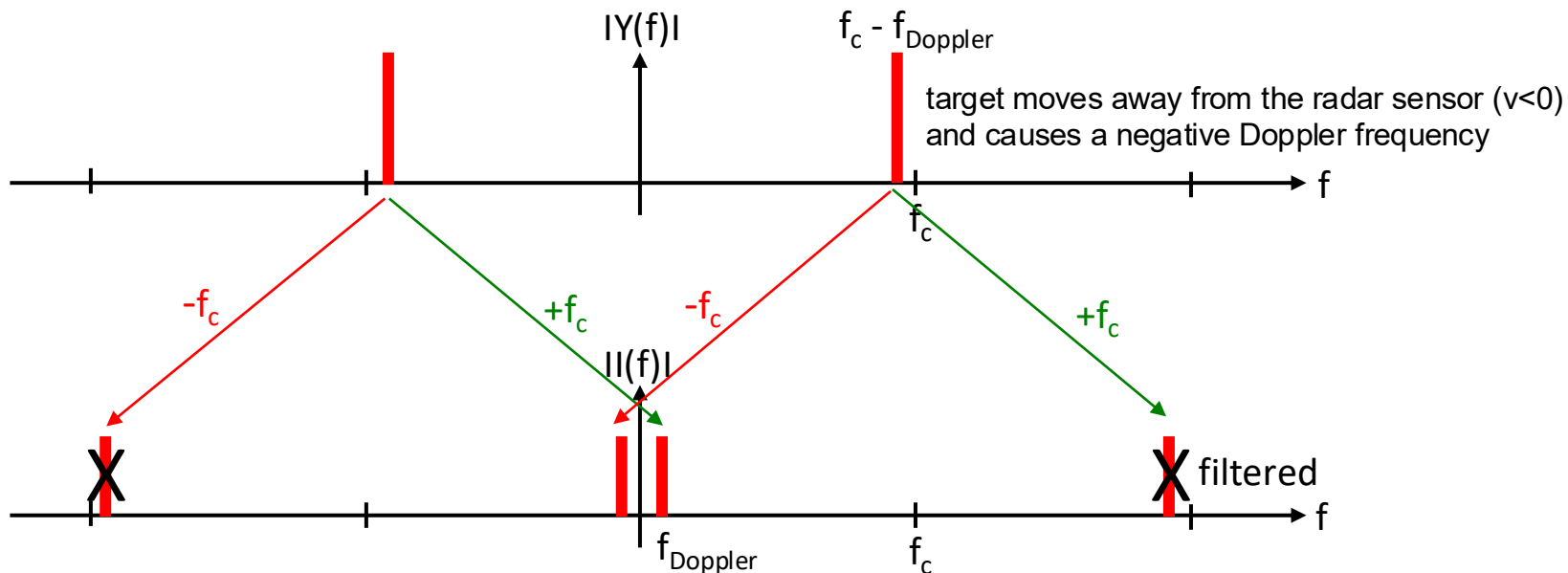
Frequency translation

$$x(t) = y(t) \cdot e^{j2\pi f_c \cdot t} \quad \circ - \bullet \quad X(f) = Y(f - f_c)$$

multiplication with $e^{j2\pi f_c \cdot t}$ $\circ - \bullet$ frequency translation with $+f_c$ (to the right)

Euler-formula: $\cos(\omega_c \cdot t) = (e^{j \cdot \omega_c \cdot t} + e^{-j \cdot \omega_c \cdot t}) / 2$

Multiplication of the received signal $y(t)$ with $\cos(2\pi \cdot f_c \cdot t)$ causes a frequency translation by $\pm f_c$ (see the following spectra)

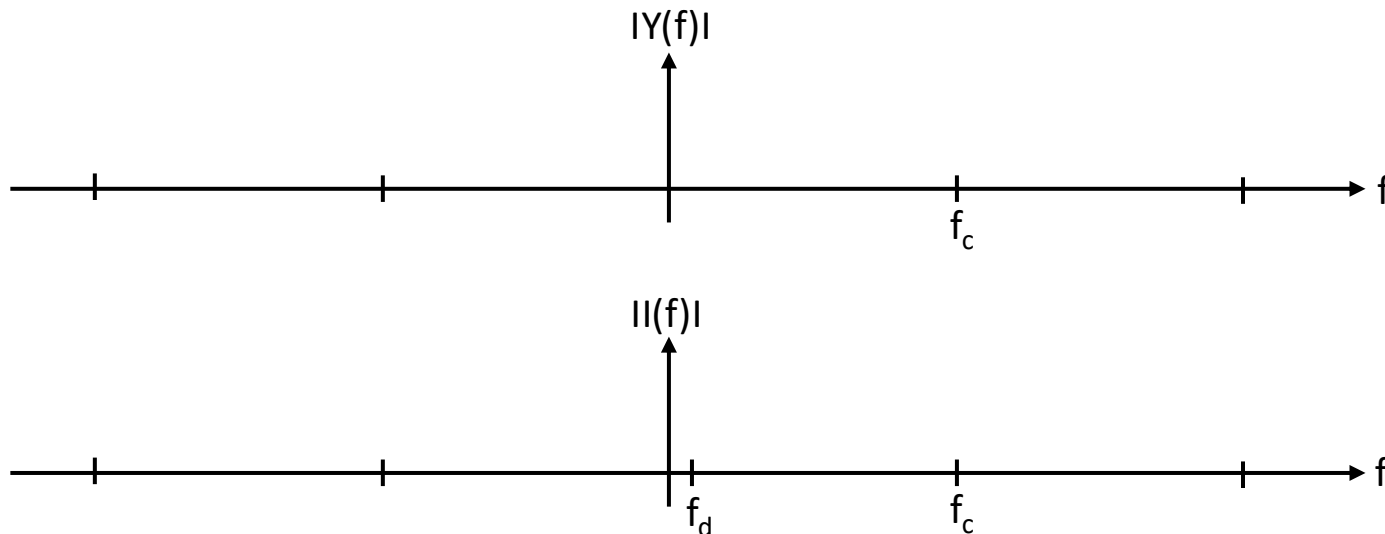


Exercise 1

Assume that the target moves towards the radar sensor ($v > 0$) and, thus, causes a positive Doppler frequency ($f_d > 0$)

In the following diagram, sketch the magnitude spectra

- $|Y(f)|$ of the reflected radar signal $y(t)$ and
- $|I(f)|$ of the analog radar-sensor output signal $I(t)$



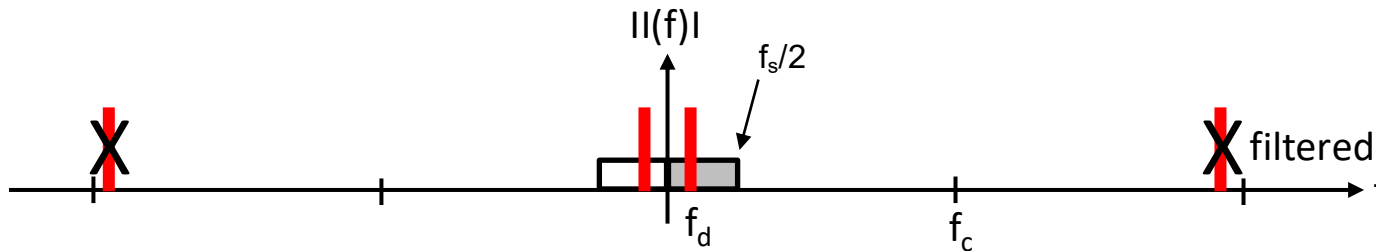
Please verify that the sign of the Doppler frequency f_d and the velocity v and, hence, the direction of the movement can not be distinguished if only 1 sensor output signal is used !

FFT-based Doppler frequency detection

Review of FFT

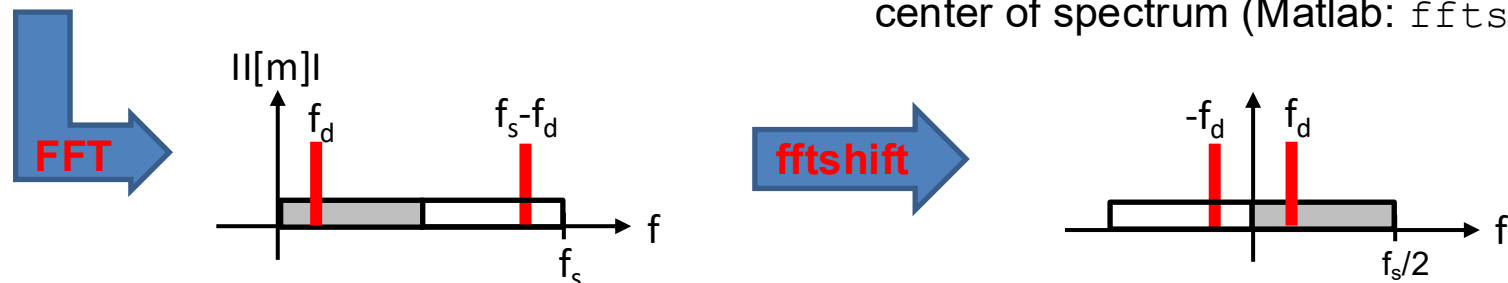
The FFT transforms N samples in the time-period $[0, \dots, NT_s)$ into N uniformly distributed (complex) spectral-values in the range $[0, \dots, f_s)$.

Spectrum of analog radar sensor output signal $I(t)$



FFT-spectrum of the sampled radar sensor output signal $I[n] = I(t=nT_s)$

(f_s is the sampling frequency)



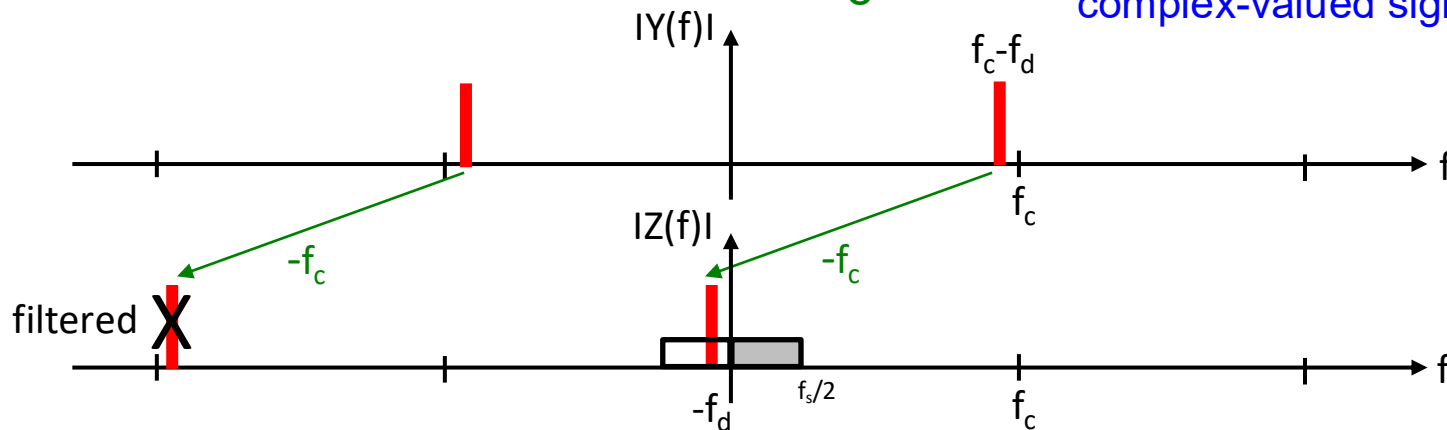
Zero-frequency component shifted to center of spectrum (Matlab: `fftshift`)

Determination of movement direction

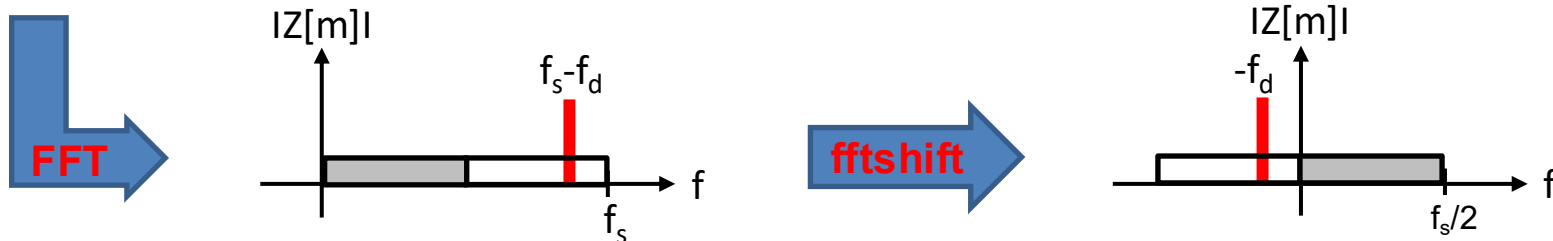
Multiplication with $e^{-j2\pi f_c \cdot t}$ ○-● frequency translation with $-f_c$ (to the left)

Euler: $e^{-j\varphi} = \cos(\varphi) - j \cdot \sin(\varphi)$

$$y(t) \cdot e^{-j2\pi f_c \cdot t} = y(t) \cdot \underbrace{[\cos(2\pi f_c \cdot t) - j \cdot \sin(2\pi f_c \cdot t)]}_{e^{-j\dots}} = \underbrace{I(t) + j \cdot Q(t)}_{\text{complex-valued signal } Z(t)}$$



FFT-spectrum of the sampled radar-sensor output signal $z[n] = I[n] + j \cdot Q[n]$

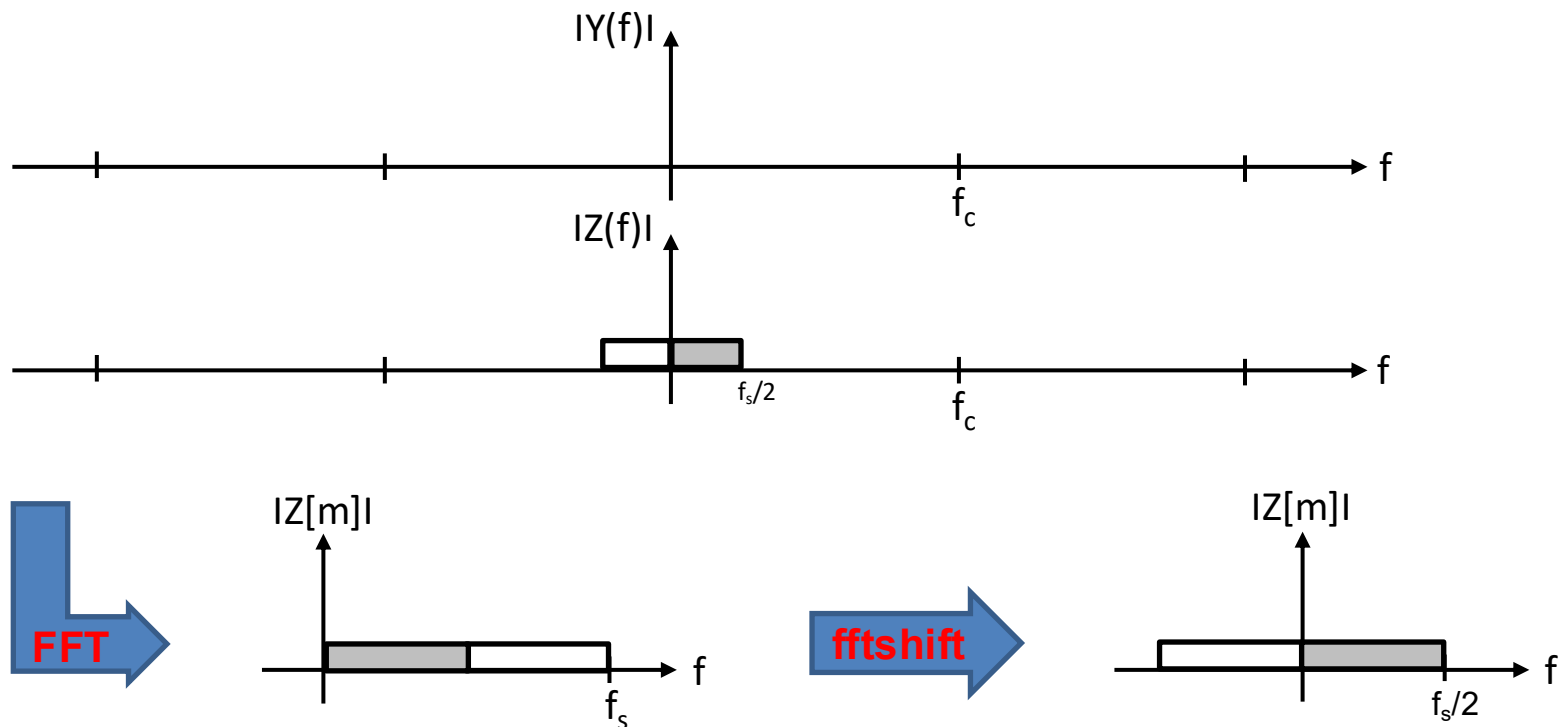


Exercise 2

Assume that the target moves towards the radar sensor ($v > 0$) and, thus, causes a positive Doppler frequency ($f_d > 0$)

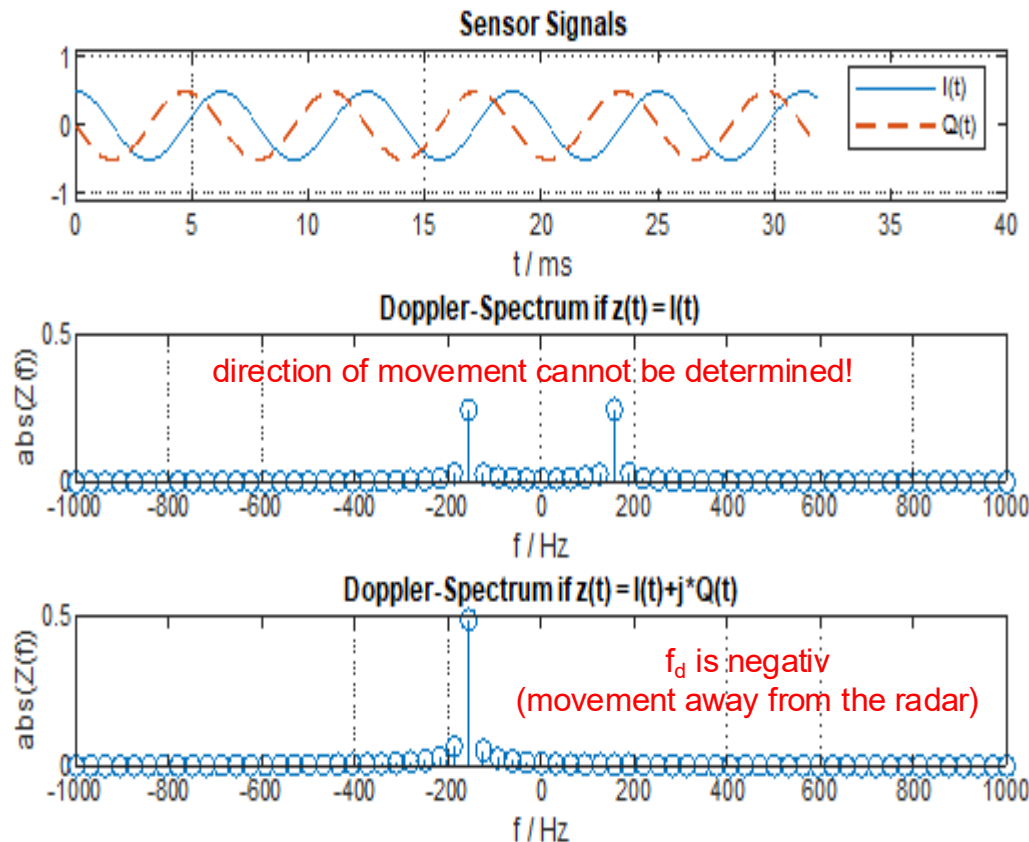
In the following diagram, sketch the magnitude spectra

- $|Y(f)|$ of the reflected radar signal $y(t)$ and
- $|Z(f)|$ of the analog radar sensor output signal $z(t) = I(t) + j \cdot Q(t)$
- $|Z[m]|$ of the sampled radar-sensor output signal $z[n] = I[n] + jQ[n]$



Exercise 3

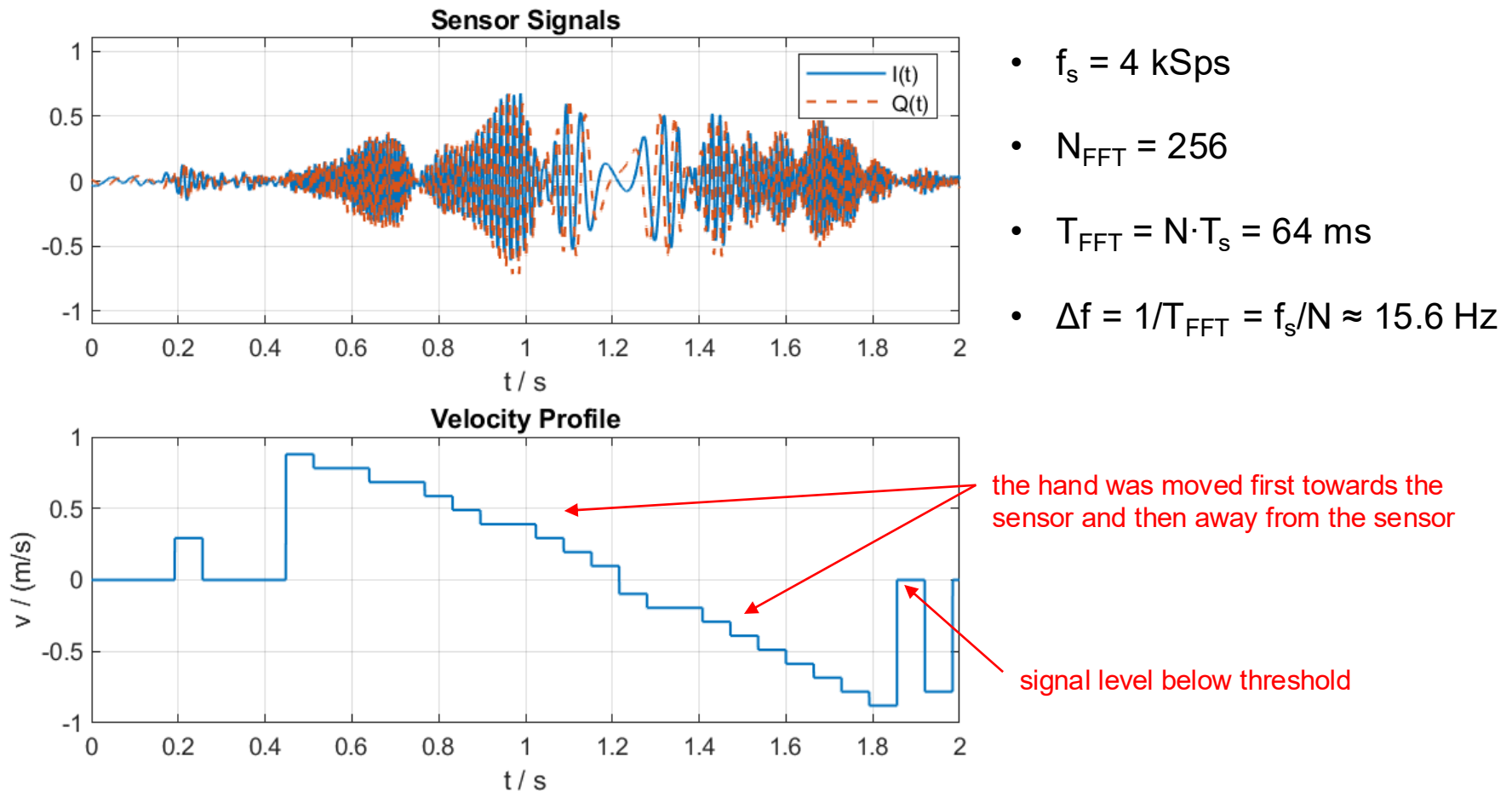
Use the Matlab-program `CW_Doppler_Radar_8kSps_160.m`, read the corresponding wave-file and determine the Doppler-frequency f_d and the resulting radial (Doppler-) velocity v . The result should look like this:



- $f_s = 8 \text{ kSps}$
- $N_{\text{FFT}} = 256$
- $T_{\text{FFT}} = N \cdot T_s = 32 \text{ ms}$
- $\Delta f = 1/T_{\text{FFT}} = f_s/N \approx 31 \text{ Hz}$
- detected $f_d = -156.3 \text{ Hz}$
(instead of -160 Hz)
- detected radial velocity
 $v = -0.98 \text{ m/s}$ (away from
the radar, instead -1 m/s)
- test also with $N_{\text{FFT}} = 64$

Exercise 4

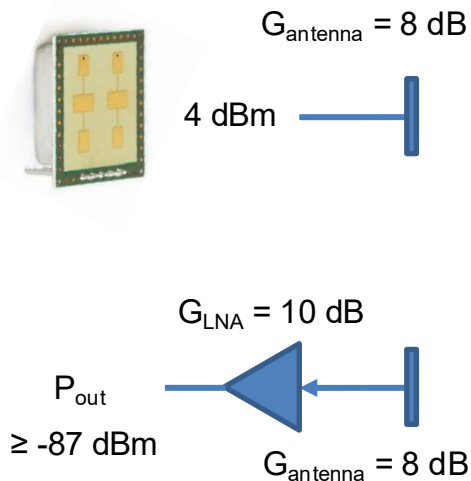
Use the Matlab-program `CW_Doppler_Radar_4kSps_waveByhand.m` read the corresponding wave-file and determine the resulting Doppler-velocity-profile $v(t)$. The result should look like this:



Speed sensor: moving car, max. range 40-60 m (see below)

- $v_{\max} = 80 \text{ km/h}$
- Doppler frequency $f_d = \pm 3556 \text{ Hz}$
- sampling frequency $f_s = 8 \text{ kSps}$ is larger than $2 \cdot |f_d|$
- block length $N = 256$
- measurement time $T_{\text{FFT}} = N \cdot T_s = 32 \text{ ms}$ (car moves $< 0.7 \text{ m}$ during this time)
- FFT frequency resolution $\Delta f = 1 / T_{\text{FFT}} \approx 31 \text{ Hz}$ (corresponds to 0.7 km/h)
- accuracy of the speed indication is $\pm 0.35 \text{ km/h}$

K-LC5 data sheet



Effective Isotropic Radiated Power

$$\text{EIRP} = 12 \text{ dBm}$$

Radar Cross Section

($\sigma \approx 1 \text{ m}^2$ for moving person
 $\sigma \approx 50 \text{ m}^2$ for moving car)

Round-trip loss $S \leq 117 \text{ dB} = 12 - (-105)$
 (Datasheet: overall sensitivity = -118 dBc)

Range R

Radar Range Equation @ 24 GHz

$$R = 0.0167 \cdot 10^{S/40} \cdot \sigma^{0.25}$$

(S in dB and σ in m^2)

Sensitivity

$$P_{\text{Rx}} \geq -105 \text{ dBm} \text{ (B=1kHz, S/N=6dB,...)}$$

Exercise 5: Compute the maximum (detection) range R of the CW-Doppler-Radar

- for a moving person:
- for a moving car:

K-LC5 data sheet

Sensitivity and Maximum Range

The values indicated here are intended to give you a 'feeling' of the attainable detection range with this module. It is not possible to define an exact RCS (radar cross section) value of real objects because reflectivity depends on many parameters. The RCS variations however influence the maximum range only by $\sqrt[4]{\sigma}$.

Maximum range for Doppler movement depends mainly on:

- **Module sensitivity**
S: -118 dBc (@ 1 kHz IF Bandwidth)
- **Carrier frequency**
 f_{TX} : 24.150 GHz
- **Radar cross section RCS "reflectivity" of the object**
 σ ¹⁾: 1 m² approx. for a moving person
> 50 m² for a moving car

The famous "Radar Equation" may be reduced for our K-band module to the following relation:

$$r = 0.0167 \cdot 10^{\frac{-S}{40}} \cdot \sqrt[4]{\sigma}$$

Using this formula, you get an indicative detection range of:

- 15 meters for a moving person.
- 40 meters for a moving car

Please note, that range values also highly depend on the performance of signal processing, environment conditions (i.e. rain, fog), housing of the module and other factors.

By reducing IF amplifier bandwidth, detection range can be enhanced. With BW = 250 Hz instead of 1 kHz, sensitivity will raise by 6 dB to -124 dBc. Maximum range will then become 21 m.

RCS

note ¹⁾ RCS indications are very inaccurate and may vary by factors of 10 and more.

Gain of external amplifier

Parameter	Value	Remarks
Sensitivity K-LC5	-105 dBm	e.g. moving person @ a range of 14 m
receiver gain	18 dB	see data sheet K-LC5
min. P_{out}	-87 dBm	@ $R = 1 \text{ k}\Omega$ output of LNA

- P_{out} in mW: $10 \cdot \log_{10}(P_{out}/1\text{mW}) \geq -87 \text{ dBm} \Rightarrow P_{out} \geq 10^{-8.7} \text{ mW}$
- $V_{out, rms}$ in V: $\sqrt{(R \cdot P_{out})} \Rightarrow V_{out, rms} \geq 10^{-8.7/2} \text{ V} = 45 \text{ }\mu\text{V} \Rightarrow V_{out} \geq 64 \text{ }\mu\text{V}$
- But: the 12 Bit ADC has a dynamic range $\approx 3\text{V}$, i.e. an LSB of $732 \text{ }\mu\text{V}$!
- therefore, an external gain amplifier is required: $V_{ADC} = V_{out} \cdot G_{amp}$
- with $G_{amp} = 60 \text{ dB}$, i.e. $G_{amp} = 1000$: $V_{ADC} \geq 64 \text{ mV}$, i.e. at least 6 LSBs
- **AC-coupling** is required for such a (high) gain, but the high-pass-filter removes the low frequencies and, thus, the small v-values.
There is a minimal detectable speed $v_{min} > 0$

Gain of external amplifier

Assumption:

a moving person is in short distance (3m) directly in front of the sensor

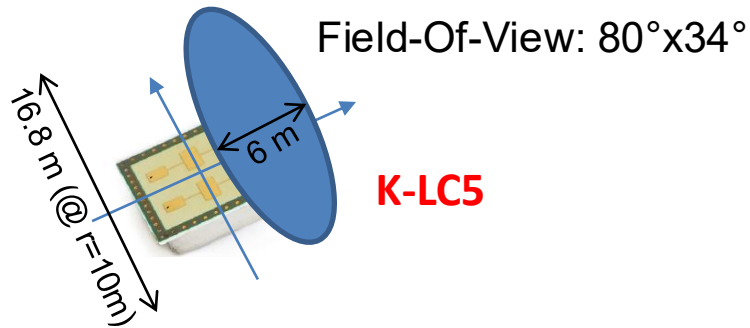
Exercise 6:

Fill out the column «values» of the following table and verify, that a «large» received signal does not over-drive the ADC (no clipping).

Parameter	Values	Remarks
EIRP	12 dBm	effective isotropic radiated power
Round-trip-loss	dB	see slide 13, $R = 3 \text{ m}$, $RCS = 1 \text{ m}^2$
received power P_{Rx}	dBm	see slide 13
receiver gain	dB	see slide 13
max. P_{out}	dBm	@ $R = 1 \text{ k}\Omega$ output of LNA
max. P_{out}	mW	@ $R = 1 \text{ k}\Omega$ output of LNA
max. V_{out}	mV	@ output of radar sensor
max. V_{ADC}	V	60 dB external gain

Radar-Sensor

[Radar Sensor K-LC5](#) [data sheet K-LC5](#)



Overall S = 118 dBc (B=1kHz, S/N=6dB, ...)

R_{max} moving person ≈ 15 m

Sensitivity: -105 dBm (B=1kHz, S/N=6dB,...)

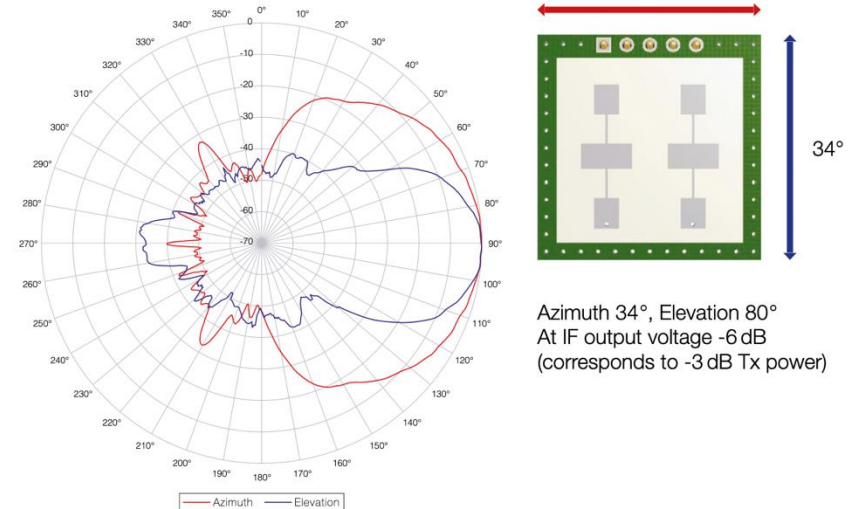
Rx gain: 18 dB

FMCW supported

3.3V version ordered

(Digikey, K-LC5-RFB-00C, CHF 25.-/pcs)

Figure 2: Antenna system diagram (logarithmic scale)



CW-Doppler-Radar: Hardware



MCU board

radar sensor K-LC5
3.3V power supply
amplifier (60 dB)
for I- and Q-output
highpass,
VCO in (DAC)

test points
LED?, switch?, buzzer?

filter analog 5V power supply
USB power supply

Sensor board

a PCB-template is available



STM32F4-Discovery-Board

Option: 3D-printed lens for focussing



Subsystem Test Scenarios

1. Sensor-Board **without** radar sensor K-LC5 and **without** MCU-board

- Function Generator (sin, 100 Hz...4 kHz, 50 μ V...1 mV) at I-/Q-connector pins
- validate the external amplifier outputs with the oscilloscope (in front of ADC)
- check also the magnitude frequency response

2. Sensor-Board **with** radar sensor K-LC5 and **without** MCU-board

- put the Radar Doppler Target K-DT1 in front of the radar sensor
- validate the external amplifier outputs with the oscilloscope (in front of ADC)



3. MCU-Board **with** FFT-firmware and **without** radar sensor board

- FuGe (sin, 100 Hz...4 kHz, 50 mV...1V, 1.5V DC-offset) at ADC inputs
- show and validate frequency on the display